

Blockchain & Ripple

Natalie, Junyu, Aadesh





**How can mutually untrusted parties
agree on a shared version of truth
without a central authority?**

“How can I know my donation does not end up lining the pockets of some corrupt local officials?”

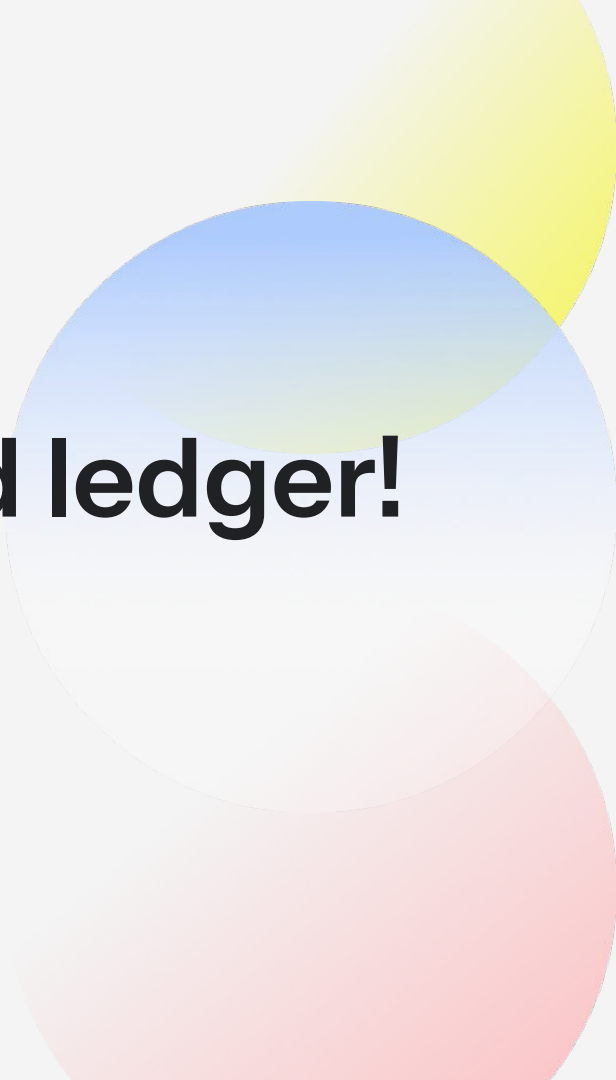
“How can I prove that I paid my rent last month if my landlord claims the opposite?”

“How can I prove that I actually graduated from USF if one day it is no more? ”

“How do I know the ‘fair trade coffee’ I bought is actually fair trade?”

Why does it matter?

- Single Point of Failure
- High cost of Cooperation/ Transaction
- Censorship/ Seizure
-



We need a shared ledger!

RippleNet in Finance



Santander

One Pay FX instant payments



SBI Holdings

60+ banks in Asia

The Real Impact

Traditional fees: **5-10%**

Ripple fees: **<1%**

Settlement: **Minutes vs. Days**

Example: Sending \$500/month US → Philippines

Annual savings: **\$300-600** back in their pocket

Live Demo



Ledger closes every
3-5 seconds



Transaction fee

~\$0.0002



Instant finality

**No confirmations
needed**

Demo link: <https://livenet.xrpl.org/>

Blockchain for Social Good

Beyond Crypto: Real Impact Happening Right Now

UN World Food Programme: Building Blocks

1M+

Refugees
Served

\$555M

Cash
Distributed

25M

Transactions
Processed

\$3.5M

Bank Fees
Saved



Humanitarian Aid

80,000+ lives

Rahat Nepal: 774 people with disabilities received aid in <6 hours during flood emergency



Land Rights

Kenya projects

Atlas Kenya & Land LayBy: Immutable land titles protect communities from fraud and corruption



Supply Chain

Every dose tracked

UNICEF Bangladesh: Blockchain vaccination tracking prevents double-counting and ensures integrity

The Pattern: Trust - Transparency - Accessibility - Censorship Resistance

What is Ripple?

Ripple ≠ XRP ≠ XRPL. They're different things.

THE COMPANY



Ripple

Private software company
(like Stripe for banks)

THE ASSET



XRP

The digital asset
(the currency)

THE CHAIN



XRPL

Open-source public
blockchain (you're here!)

You're building on the XRPL — not Ripple. Like using the internet, not Google.

The Universal Translator

Why does this tech even exist?

THE OLD WAY

Bank A

Correspondent Bank

Central Bank

Bank B

3–5 days



Time

\$30+



Cost

THE XRPL WAY

Bank A



XRPL (XRP bridge)



Bank B

3 seconds



Time

\$0.0001



Cost

No New Languages Required

HTTP POST request? You're already an XRPL developer.

• • • xrp1_payment.json

```
{  
  "TransactionType": "Payment",  
  "Account": "rHb9CJAWb...",  
  "Amount": "1000000",  
  "Destination": "rPT1Sjq...",  
  "Fee": "12"  
}
```

It's just a Web API. That's it.

Works with your stack

JavaScript / Node.js

Python

Java

Go



Solidity / Rust / Vyper

Your Hackathon Stack

Demystifying the 'blockchain' part. Spoiler: it's just an API call.

LAYER 1

1



Frontend UI

React • Vue • Mobile

LAYER 2

2



Backend Logic

Node.js • Python
AWS Lambda

LAYER 3

3



XRPL Testnet

WebSocket / REST
API Endpoint

Think of XRPL as swapping your PostgreSQL call for a decentralized API endpoint. Same pattern.

Native Lego Blocks

No need to write token or escrow logic from scratch. It's baked right in. (actual API functions)



TrustSet

Custom Token Maker

Reward users with your own token for organizing real-life activities



EscrowCreate

Digital Trust Fund

Lock charity funds until a milestone condition is met



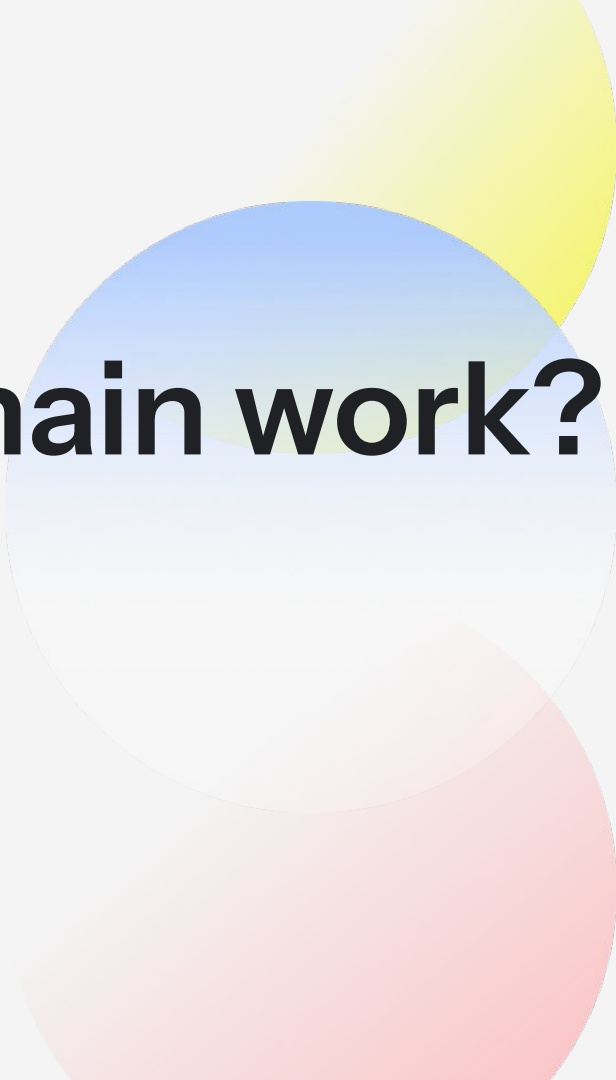
NFTokenMint

Digital Passports

Create digital twins for physical items with attached metadata

On other chains you'd write this from scratch. On XRPL, just pass parameters.

How does Blockchain work?



Hash: A fingerprint generator

Input

“Mary had a
little lamb”

“Mary ate a
little lamb”

Hash Function

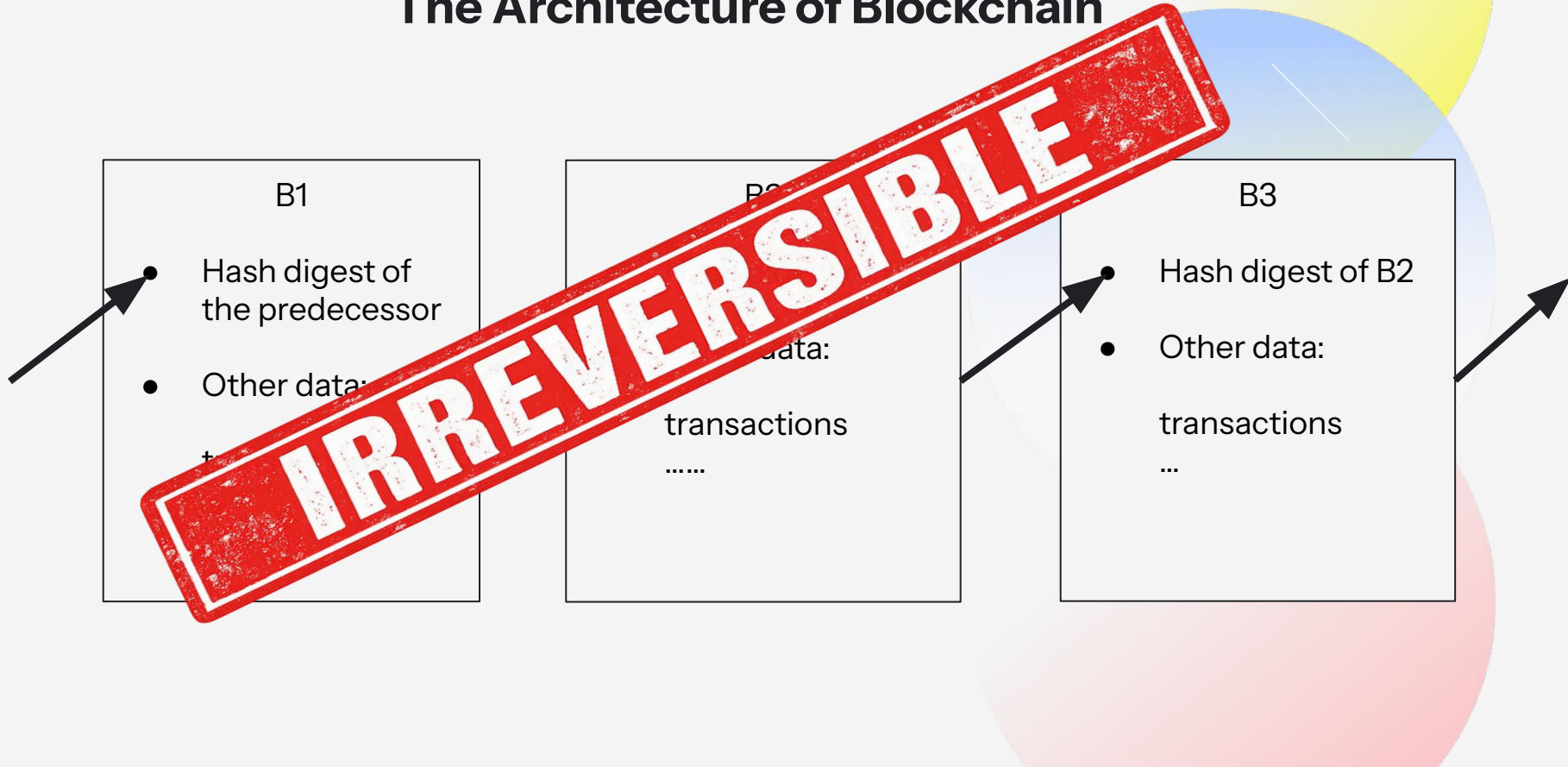
Hash Digest

“15e2b0d3c33891ebb0f1ef60
9ec419420c20e320ce94c65f
bc8c3312448eb225”

“9f3c7a8b1d6e42c5e8a1f4b97
c3d2a5e6f8b0c1d4e7a9b2c3d
6f1a8e5b4c7d9”

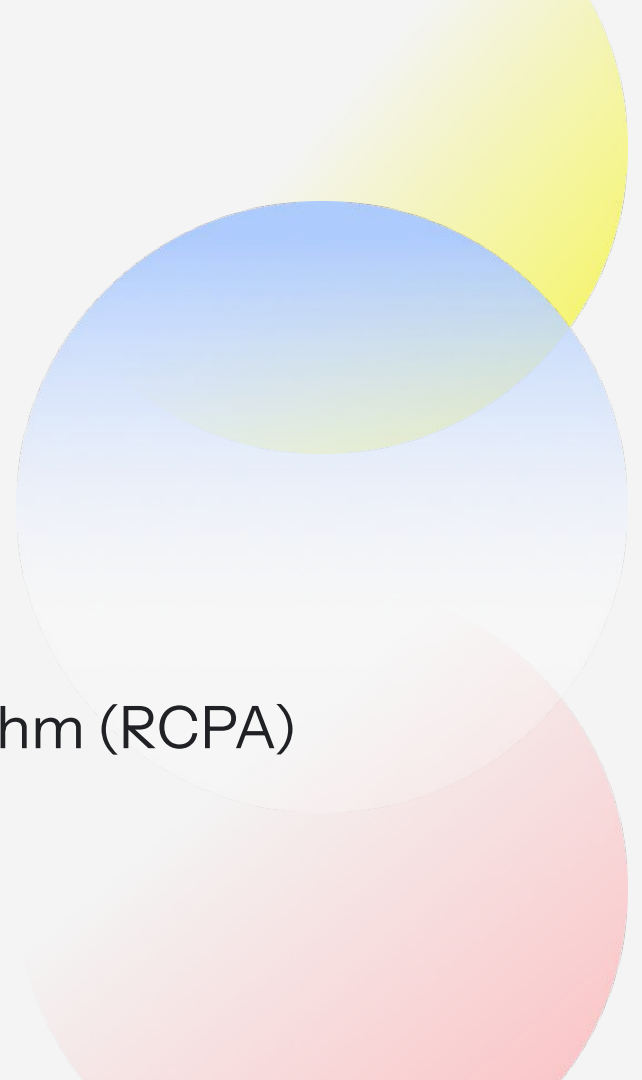


The Architecture of Blockchain



Consensus

- Proof of Work (PoS)
- Proof of Stake (PoS)
- Ripple Protocol Consensus Algorithm (RCPA)
-



Resources to Get Started

XRP Ledger: livenet.xrpl.org + xrpl.org/docs

News: [CoinDesk](#), [Ripple Insights](#)

Learn to code on the XRP Ledger: <https://learn.xrpl.org/>



Scan Me